

Complementary Events

Q- How do I calculate the probability of complementary events?

Questions-

What does complement mean when in probability?

What does the probability of an event and its complement add up to?

How do I use this?

Notes-

① The complement of an event E is written E' . E' is the event that E does not occur.

e.g

The complement of rolling a 3 is not rolling a 3.

$$\Pr(\text{red}') = \Pr(\text{not getting red})$$

$$\Pr(E) + \Pr(E') = 1$$

e.g $\Pr(\text{rolling a 5}) = \frac{1}{6}$

$$\Pr(\text{not rolling a 5}) = \frac{5}{6}$$

If the probability of choosing a red marble from a bag is $\frac{2}{7}$. The

probability of not choosing a 5 is $\frac{5}{7}$

If $\Pr(\text{blue}) = 0.3$ then $\Pr(\text{not blue}) = 0.7$

② Summary- The complement of an event is the event not happening. The probability of an event and its complement need to add up to 1. If $\Pr(E) = \frac{3}{5}$ $\Pr(E') = \frac{2}{5}$ or as a decimal $P(E) = 0.4$ then $P(E') = 0.6$.